



Operators in Python

Practical Skills Module 3: Topic 2

[Visit our website](#)

Practical Skills Module 3: Topic 2

Topic Code	PM-03-PS02
Topic	Use Operators

Scope of Practical Skill

Given a range the learner must be able to:

- PA0201 Perform various arithmetic calculations
- PA0202 Compare values and print the result
- PA0203 Assign values
- PA0204 Combine conditional statements and compare (binary) numbers
- PA0205 Test for membership in a sequence such as lists, strings or tuples
- PA0206 Compare the memory location of two objects
- PA0207 Set priority for which calculation need to be done first in a complex calculation
- PA0208 Use bitwise operators

Applied Knowledge

- AK0201 Concept, definition and functions of Python programming functionalities

Internal Assessment Criteria

- IAC0201 Expected results with Python operators are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-03



Practical task

Follow the facilitator's instructions to complete the following activities:

Create a Python file named **student_grading.py** to implement the following tasks and ensure it is stored in your task folder.

Scenario: Student Grading and Performance Analysis

A school wants to develop a Python application to help teachers and administrators evaluate student performance. This system will calculate academic scores, assign grades, determine award eligibility, and perform other evaluations and checks to assist in student analysis.

Step 1: Perform Basic Arithmetic Calculations

Begin by calculating a student's:

- Total marks
- Average marks
- Percentage

Use the following marks as an example:

- Math: 85
- Science: 90
- English: 78

Step 2: Determine the Grade Based on Percentage

Use conditional statements to compare the calculated percentage with grade thresholds and print the appropriate grade:

- Grade A: percentage ≥ 90
- Grade B: percentage ≥ 75 and < 90
- Grade C: anything below 75

Step 3: Assign Award Eligibility

Based on the percentage or grade:

- Eligible for an award: percentage ≥ 90
- Not Eligible: otherwise

Print a message indicating the student's eligibility.

Step 4: Combine Multiple Conditions Using Logical Operators

Check if the student scored 80 or above in all three subjects. If so, print a message indicating strong subject-wise performance.

Step 5: Check for Membership in a Sequence

Create a list of top-performing students:

```
top_students = ["Alice", "Bob", "Charlie"]
```

Check whether the current student's name (e.g. "Alice") is in this list, and print an appropriate message.

Step 6: Compare the Memory Locations of Two Grade Variables

Create two variables:

```
grade1 = "A"
```

```
grade2 = "A"
```

Use the `is` operator to check whether both variables refer to the same memory location, and print the result.

Step 7: Perform Complex Weighted Score Calculations

Calculate a weighted score using specific weights for each subject:

- Math = 40%
- Science = 35%
- English = 25%

Make sure to use parentheses to control the order of operations and print the final weighted score.

Step 8: Use Bitwise Operators for Student ID Checks

Assign each student a unique integer ID (e.g. using binary values). Use bitwise operations (&, |, ^, etc.) to simulate checking for shared properties, such as course enrolments or flags. Print the results to demonstrate how these operators work.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
